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Palaeomagnetism of a Caledonian ophiolite complex, Karmøy, southwest Norway

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The palaeomagnetism of the Karmøy ophiolite suite emplaced at the southwest extremity of the Scandinavian Caledonides in Middle-Upper Ordovician times is described. Low intensities of magnetisation and low Q_n values in the bulk of the constituent facies are linked to the effects of low grade metamorphism, but ultramafic pods that have experienced serpentinisation yield well-defined single component magnetisations. A predominant (A) component of magnetisation isolated in 35 samples yields a mean of $D/I = 51°/58°$ ($\alpha_{95} = 5°$). This population is discrete from a subsidiary (B) component recognised in 17 samples with a mean direction of $145°/63°$ ($\alpha_{95} = 8$); a minor (C) component ($D/I = 330°/60°$, $\alpha_{95} = 8°$) is also identified and small numbers of anti-parallel components are identified in each case. An Upper Ordovician ignimbrite from Kattnakken, northeast of Karmøy, yields an in situ component of $D/I = 270°/38°$; together with similar components recognised elsewhere, this mean ('D') direction is most closely compatible with directions recognised in Upper Silurian–Lower Devonian rocks from the Scottish sector of the Caledonides; it is interpreted as an overprint of this age, but may include a component of local rotation.

Two interpretations are considered for the remanence record in the Karmøy complex. The 'A' component could record the influence of a thermal event linked to uplift of the continental margin and development of the adjoining North Sea Basin in Mesozoic times if this margin has been tilted seawards by ~ 20° since magnetisation. No plausible tectonic model can bring the 'B' component into conformity with the post-300 Ma apparent polar wander (APW) path and it is alternatively proposed that both the 'A' and 'B' components record Upper Ordovician post-emplacement remanences. With this interpretation the data complement a range of palaeomagnetic results from Middle–Upper Ordovician igneous and metamorphic rocks from North Atlantic bordering terranes yielding intermediate to high palaeolatitudes and contrasting with older and younger data. Collectively they imply that the Iapetus Ocean had effectively closed by these times, and that large APW movements took place between mid-Ordovician and mid-Silurian times.

1. Introduction

Attempts to quantify the closure of the Lower Palaeozoic Iapetus Ocean, which culminated in the Caledonian orogenic episode, have currently only met with limited success and have mostly involved palaeomagnetic study of terranes now sited within the orogen rather than from its opposing margins (Deutsch, 1980; Briden et al., 1984; Piper, 1987; Trench et al., 1988). The evidence for this former ocean basin is recorded by the linear distribution of ophiolite wedges along the orogen that comprise remanents of the former ocean crust and are mostly of Ordovician age (Dewey, 1969; Sturt and Thon, 1978a). These obducted bodies are of little use as palaeomagnetic indicators of ocean closure because their emplacement has typically been accompanied by deformation and pervasive greenschist-facies metamorphism. Most studies have found that the pillow lava, sheeted dyke and gabbro members of the pseudostratigraphy have had their magnetism largely, or com-

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pletely, obliterated by the effects of low grade metamorphism (e.g. Taylor, 1988). However, the ultrabasic facies have often not undergone the same kind of mineralogical changes leading to resorbtion or recrystallisation of the ferrimagnetic constituents; instead new magnetite growth associated with serpentinisation has caused them to acquire a stable and coherent magnetisation of palaeomagnetic significance (Swift and Johnson, 1984; Piper, 1987; Taylor, 1988).

In Norway, ophiolite bodies have been emplaced along the full ~1200 km length of the Scandinavian Caledonides. At the extreme southwest tip of this belt the Karmøy and associated ophiolites on Bømlo and Stord (Sturt and Thon, 1978a) are overthrust on to the southeast foreland of the orogen. The present paper reports a palaeomagnetic investigation of the Karmøy ophiolite together with results from some related rocks from the northeast strike continuation within this sector of the Caledonides. It is relevant both as a contribution to the at present poorly understood remanence record in the Scandinavian Caledonides, and because this body is in closest proximity to the southwest continuation of the orogenic belt in the British Isles. In this same sector of the belt the Unst ophiolite of Shetland on the opposite side of the North Sea Basin is overthrust on to the opposite northwest foreland of the orogen. It has yielded palaeomagnetic results that are only in part interpretable in the context of the Caledonian and younger apparent polar wander (APW) path for Europe (Taylor, 1988); this study of Karmøy provides a comparative investigation.

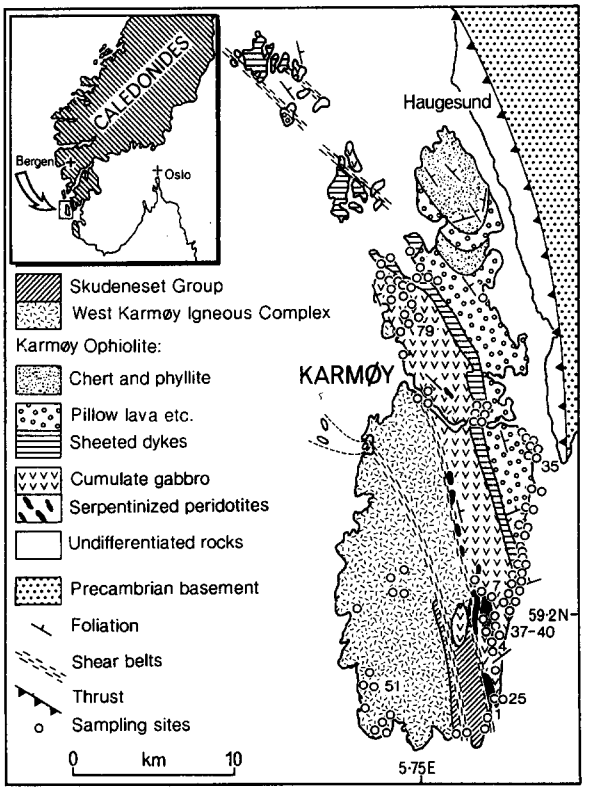


Fig. 1. Geological map of the Karmøy ophiolite complex after Sturt et al. (1979) with sites sampled for this palaeomagnetic investigation. These sites are too numerous to number, but sites yielding significant directional data are numbered and all components belonging to the 'A' and 'B' groups are derived from the southeast of the complex.

2. Geology

The lowest member of the Karmøy ophiolite (Fig. 1) comprises layered gabbros passing upwards into a pegmatitic facies and ultimately into isotropic gabbro (Sturt et al., 1979, 1980). This unit records numerous episodes of emplacement and deformation although the rocks are uniformly altered to aggregates of epidote, clinozoisite and actinolite. The lowest part of the gabbro incorporates serpentinised ultramafic lenses that include cumulate layered bodies, narrow dykes of peridotite, and bodies that may represent detached fragments of upper mantle peridotite (Sturt et al., 1979). The gabbros are overlain by a fairly complete and extensive sheeted dyke complex about 2 km thick, and poorly represented pillow lavas and breccias originating in an abyssal environment (Sturt et al., 1980). As in the case of the gabbros, all primary minerals have been converted to low grade assemblages. The presence of an overlying greywacke suite (the Visnes Group) and the abundance of brecciated mass flow deposits suggest that the ophiolite originated near to a ridge–ridge transform system; this is also implied by geochemical evidence (Furnes et al., 1980).

The present outcrop of the Karmøy ophiolite (Fig. 1) comprises the southwest limb of a major syncline that plunges to the northwest and is associated with an axial planar cleavage that was probably acquired during the obduction of the ophiolite (Sturt et al., 1980). The thrust contact with the adjoining Fennoscandian Shield is exposed on the Haugesund Peninsula immediately to the east of Karmøy (Fig. 1).

The base of the complex is now obscured by a suite of silicic intrusions ranging in composition from quartz diorite to granite and grouped as the West Karmøy Igneous Complex (referred to hereafter as WKIC; Sturt and Thon, 1978b). This has an emplacement age post-dating the polyphase deformation and greenschist-facies metamorphism of the ophiolite. A provisional Upper Ordovician (Ashgillian) minimum age of 445 Ma (data of Raheim quoted in Sturt et al., 1979, 1980) is based on Rb–Sr whole-rock determinations from late granodiorite dykes. Petrologic evidence and the high $^{87}Sr/^{86}Sr$ ratios in the WKIC imply a derivation by partial melting of continental crust accompanying ophiolite obduction (Furnes et al., 1980; Sturt et al., 1980). The intrusion was unroofed prior to deposition of a transgressive clastic marine sequence, the Skudeneset Group, containing a shallow marine fauna also of late Ordovician (Ashgillian) age (Broch et al., 1940) and defining a minimum stratigraphic age for the events outlined above. During the late Silurian stage (Main Scandinavian Phase; Roberts and Gale, 1978) of orogenic deformation, the basement incorporating the ophiolite and WKIC was cut by a few widely spaced, steep and linear, NNW–SSE trending mylonite zones (Fig. 1); they have accommodated simple shear with sinistral movements and uplift to the east (Thon, 1980). Local metamorphic retrogression was also associated with this episode (Thon, 1980).

3. Palaeomagnetic results

One hundred sites were sampled in all units of the ophiolite complex on Karmøy (Fig. 1), with the exception of the cherts and associated units in the northern part of the island. The sample includes a collection from the WKIC and associated xenoliths, in addition to mélange units of the Skudeneset Group. These sites yielded a total of 778 samples collected as cores using a portable drill. Samples were oriented by standard methods employing sights on the Sun and/or magnetic bearings in conjunction with topographic sights. In the laboratory, cores were progressively demagnetised following standard thermal and a.f. cleaning techniques and measured by parastatic magnetometers; a collection of weakly magnetised material was measured by a cryogenic magnetometer incorporating a progressive a.f. cleaning facility (Shaw et al., 1984).

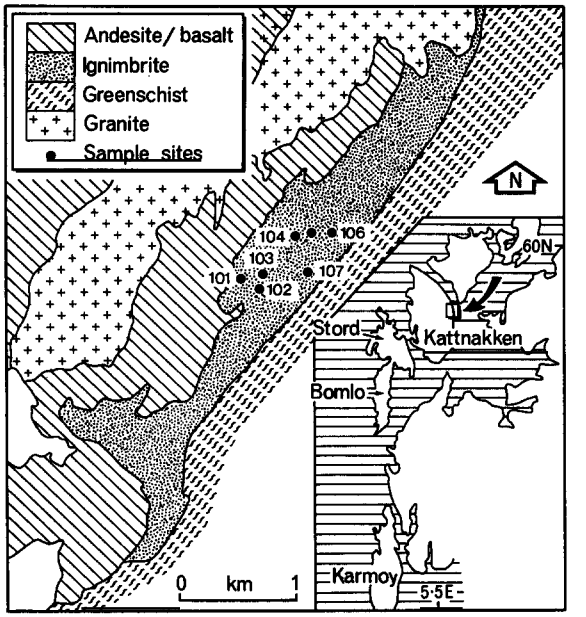


Fig. 2. Simplified geological map of the Stord area, southwest Norway (after Furnes and Thon, 1981) illustrating palaeomagnetic sampling sites in the Kattnakken ignimbrite.

On the island of Stord, 40 km to the north-northeast of Karmøy, an ophiolite succession, which is probably correlative with part of the Karmøy suite, is in tectonic contact with a varied volcanic suite (the Hardangerfjorden Group). It includes an ignimbrite unit with a vertical thickness of some 600 m, which is now overturned and exposed on the inverted limb of a NE–SW anticline at Kattnakken (Furnes and Thon, 1981; Fig. 2). The ignimbrite has yielded a Rb–Sr age of

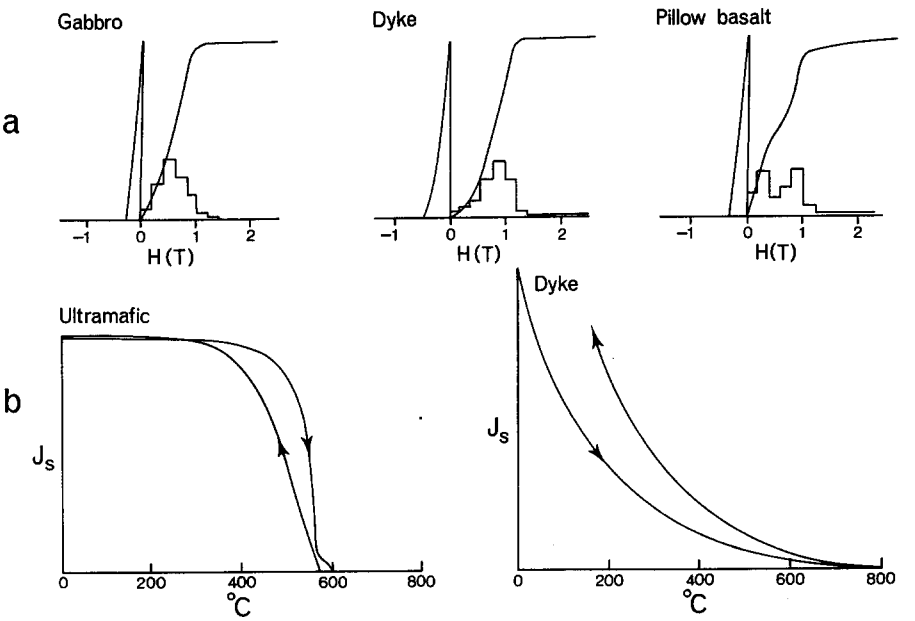


Fig. 3. (a) Representative IRM spectra for examples of the ophiolite facies. H is the magnetising field in Tesla and the induced magnetisation axis is arbitrary. Note that these samples are saturated in low fields typical of magnetite or sulphide carriers. (b) Thermomagnetic (saturation magnetisation, J_s, versus temperature) curves for samples of the ophiolite illustrating discrete magnetite and a dominance by paramagnetic constituents.

445 ± 5 Ma (Lippard and Mitchell, 1980) similar to the WKIC, and is probably older than the nearby Dyvikvagen Group containing an Ashgill to Lower Llandovery fauna. Our palaeomagnetic sample from Stord comprises 46 samples from seven sites through the ignimbrite succession (at 5.5° E, 59.9° N, Fig. 2) and four sites from the ophiolite suite (at 5.4° E, 59.8° N). These sites were collected either as drilled cores or as oriented blocks for laboratory coring.

The main feature of the magnetic properties of this collection is the dominance of low coercivity and low blocking temperature components in all facies except the ultramafics. This is illustrated by induced remanent magnetisation (IRM) studies (Fig. 3a), which identify fairly low saturation fields but are not specifically diagnostic of multidomain or pseudo-single domain magnetite carriers. The thermomagnetic curves for these samples are typically exponential and illustrate a dominance of the saturation remanence (J_s) by paramagnetic phases obeying the Curie Law (Fig. 3b). Low temperature susceptibility studies (Fig. 4) show trends similar to type 1 of Senenayake and McElhinny (1981) and linked by them to samples with a susceptibility dominated by homogeneous multidomain magnetite; these samples exhibited unstable behaviour during progressive cleaning. Ultramafic samples and a few dykes illustrated susceptibility behaviour similar to type 3 with a peak at about

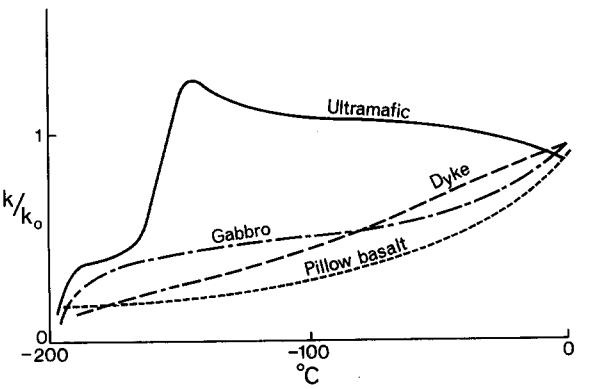


Fig. 4. Representative low-temperature susceptibility determinations from members of the ophiolite suite.

−140 °C due to a change in sign of the principal anisotropy constant; it is probably indicative of grain subdivision by exsolution of a Ti-rich component (Fig. 4) with the remanence-carrying phase being near to pure magnetite in composition (Fig. 3b). These samples show stable behaviour and yield statistically significant natural remanent magnetisation (NRM) components during progressive cleaning; we attribute their strong magnetic properties to the effects of serpentinisation (Sturt et al., 1979).

Only 94 samples from Karmøy (12% of the total measured collection), which come mainly from nine sites in ultrabasic bodies, yielded remanence components defined by at least three successive demagnetisation steps. The poor stability of the remaining collection is reflected in the low Q_n ratios, which distinguish the basaltic to gabbroic facies of the ophiolite from the ultramafic facies (Fig. 5). This probably reflects a thorough conversion of primary igneous minerals in the lavas, dykes and gabbros to low grade metamorphic assemblages with associated incorporation of the iron into metamorphic amphiboles and chlorite. The magnetic consequence of this low grade metamorphism is a reduction in remanent intensity (J_0) and a consequent reduction in Q_n value. This is evident from the observation that the shift towards the fields of lowest susceptibilities and intensities is most marked in the fine-grained facies of the ophiolite, and least in the coarse-grained facies (Fig. 5).

All components meeting the acceptibility criteria (defined by three or more steps of treatment yielding a mean vector with a half angle of cone of 95% confidence $< 20°$) come from samples of the ultramafic pods in the gabbros, with the exception of a few samples from the other ophiolite facies and four from the WKIC. The majority of stable samples show single component behaviour with a predominance of northeast and southeast declinations and intermediate to steep inclinations (see Fig. 9). Comparable results are yielded by both a.f. and thermal cleaning and there is no significant difference between the vector directions isolated by each method. The most prominent population of directions has a northeast declination and intermediate inclination (see Fig. 9). Directions belonging to this ('A') group are defined by distributed coercivity and blocking temperature spectra, and are unblocked close to the Curie point of magnetite. These magnetisations have declinations ranging from eastnortheast to northnortheast and are essentially single component (samples 1-5, 4-2, 37-7, 37-9 and 39-1 in Fig. 6); in one sample (37-4 in Fig. 6) the predominant eastnortheast positive direction is isolated as a magnetite-held component with a small residual component of opposite polarity identified above 580 °C and presumably resident in haematite.

A subordinate and more diffuse grouping of components has comparable inclinations but southeast declinations. These are assigned to a 'B' group and representative examples are illustrated in Fig. 7. They are also unblocked close to the Curie point of magnetite (e.g., sample 78-3 in Fig. 7). They tend to have somewhat higher coercivities than the 'A' group (cf. 7-1 and 25-7 in Fig. 7 and 37-7 in Fig. 6), although a few samples show the 'B' component present as a lower blocking temperature/coercivity component and the 'A' component as the higher blocking temperature/coercivity component (78-3 and 4-10 in Fig. 7). Although remanence components were defined by the weakest fraction of the collection by measurement with cryogenic magnetometers, they had a random distribution; as in the case of the Unst Complex on Shetland (G.K. Taylor, personal communication), no additional useful information has been forthcoming from the fraction of the collection with intensities-below the sensitivity of the parastatic magnetometers ($\sim 0.5 \times 10^{-7}$ A m^{-2} kg^{-1}). No other significant components show any apparent groupings (see Fig. 9) with the exception of 10 northnorthwest-directed magnetisations. Sample 40-9 in Fig. 8 is an example of one of these components.

Only one sample from the ophiolite suite on the island of Stord yielded a significant characteristic NRM component ($D/I = 199°/75°$ over a blocking temperature range 0–500 °C with thermal treatment). The Kattnakken ignimbrites, however, proved to be more suitable for palaeomagnetic investigation, with 16 samples from the seven sites (Fig. 2) yielding significant components. The

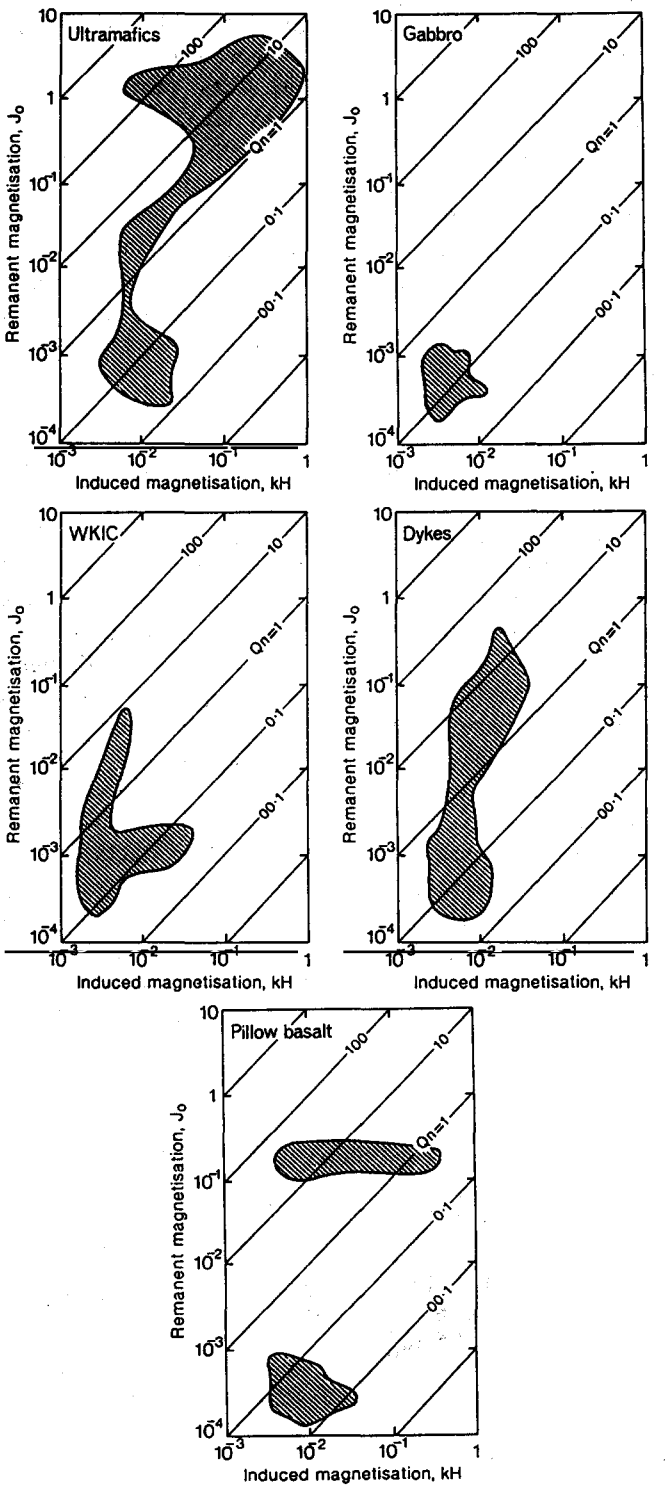

Ultramafics
Gabbro
WKIC
Dykes
Pillow basalt
Remanent magnetisation, Jo
Induced magnetisation, kH
Qn=1
100
10
0·1
0·01
10
1
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻³
10⁻²
10⁻¹
1

TABLE 1

Palaeomagnetic results, Karmøy Ophiolite and related rocks, southwest Norway [a]

Pole reference	N	D	I	α_{95}	k	Palaeomagnetic Pole		
						°E	°N	dp/dm
Karmøy Ophiolite								
A	35 [b]	50.6	57.6	4.8	26.7	105.8	51.9	5.1/7.0
B	17 [b]	144.7	62.5	7.1	26.2	31.2	17.1	8.7/11.1
C	10 [b]	329.7	59.9	7.8	39.2	244.7	63.6	8.9/11.8
Stord (Kattnakken)								
	9	269.9	37.7	8.2	40.2	286.7	18.1	5.7/9.7
D	19 [c]	261.1	46.2	8.9	15.1	297.8	19.1	7.3/11.4

[a] D and I are the mean declination and inclination of the remanence vector derived from N samples; k is the estimate of Fisher's precision parameter ($k = (N-1)/(N-R)$; α_{95} is the semi-angle of the cone of 95% confidence about the mean direction.
[b] Mixed polarities.
[c] Incorporating comparable components from Karmøy.

majority of these show single component behaviour above 10 mT (Fig. 8), with westerly declinations and shallow positive inclinations. Thermomagnetic, IRM and demagnetisation behaviour experiments on these ignimbrites suggest that the remanence carrier is magnetite.

Largely absent within the Karmøy ophiolite are southerly shallow directions of magnetisation usually regarded as typical of the Palaeozoic palaeomagnetic record from Europe. Only five or six sample directions could reasonably be assigned this origin (Fig. 9). The 'A' population is defined by a dual-polarity axis and was evidently acquired during a protracted event embracing at least one inversion of the geomagnetic field. This majority group yields an overall mean of $D = 51°$, $I = 58°$ with a derived palaeopole (A) at 106°E, 52°N (dp/dm = 5.1°/7.0°, Table 1). The group of 16 southeast, steep positive components with one antiparallel equivalent yields a mean direction of $D = 145°$, $I = 63°$ (Table 1 and Fig. 9) and an equivalent palaeopole (B) at 32°E, 17°N. The 'A' and 'B' directions are all isolated from rock units within the southeast extremity of the complex (Fig. 1); they sometimes occur at the same site and no simple distinction can be made between samples falling into the two groups. As noted above, samples falling in the 'B' group can have both higher and lower coercivities and blocking temperatures than the samples falling in the 'A' group. A further group of seven northwest positive components with three antiparallel directions yields a mean of $D = 330°$, $I = 60°$ and a palaeopole (C) at 245°E, 64°N. Only one group of palaeomagnetic directions (westerly positive) is evident from the Kattnakken ignimbrites on Stord. Unfortunately the outcrop of this unit is dipping nearly uniformly at ~ 40° to the northwest, and a fold test to determine the age of this remanence with respect to the regional folding is not effective.

Fig. 5. The distributions of induced magnetisation versus remanent magnetisation for palaeomagnetic samples from the Karmøy ophiolite. $Q_n = J_0/kH$ where J_0 is the total remanent intensity, k is the mass susceptibility and H is the ambient magnetic field. Samples characterised by Q_n ratios greater than ~ 0.5 tend to yield components of magnetisation stable to progressive cleaning treatment, whereas samples with Q_n less than this value are usually unstable (Stacey, 1963). The distributions include results from 29 gabbros, 35 pillow basalts, 116 dykes, 28 samples of granites and xenoliths from the West Karmøy Igneous Complex and 22 samples from ultramafic pods. Note that the majority of pillow lava samples have very low Q_n values, although seven samples have high values and most of these yield stable components of magnetisation.

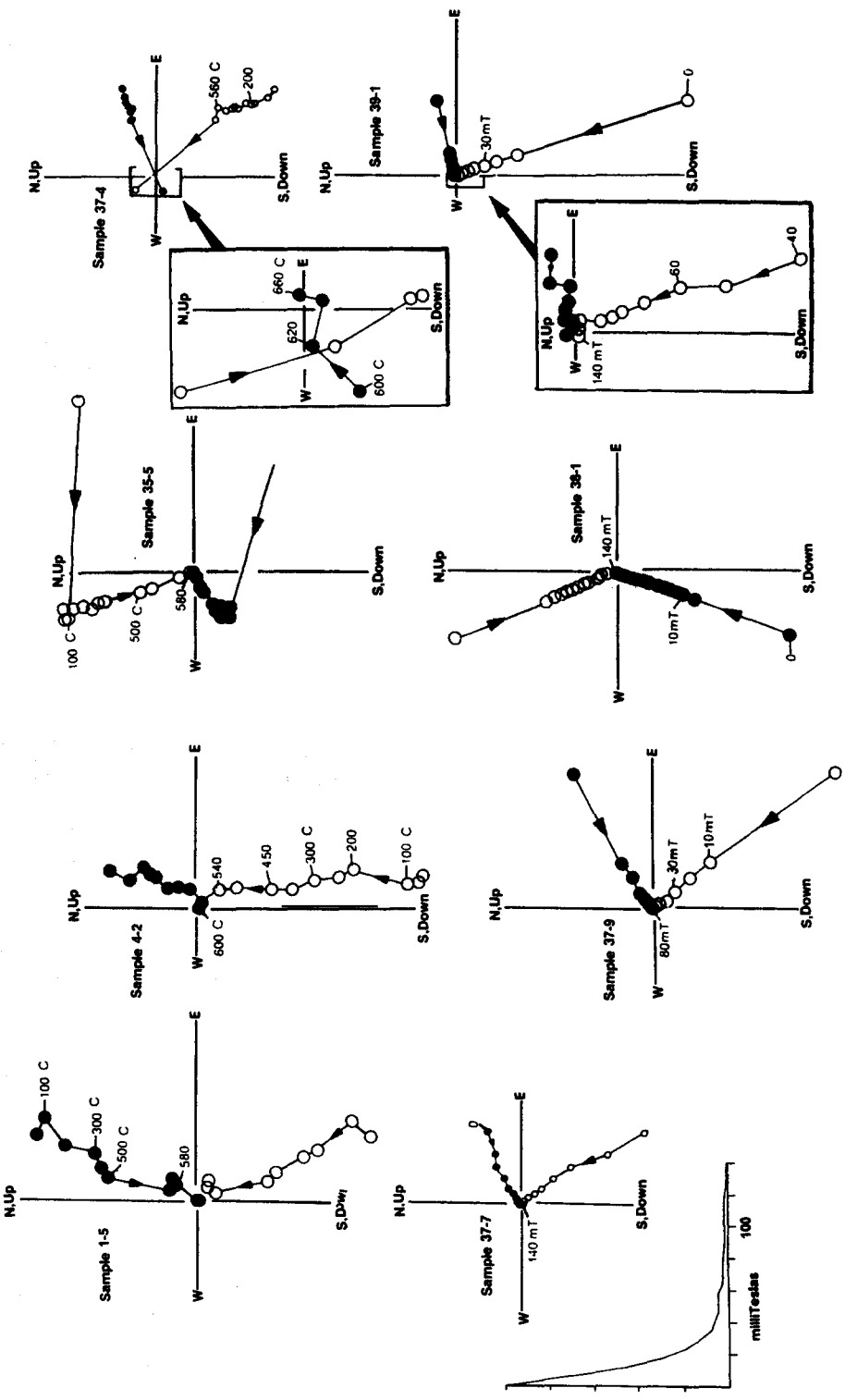


Fig. 6. Orthogonal vector plots illustrating 'A' (NE+) group components of magnetisation resolved by thermal and alternating field cleaning of samples from the Karmøy ophiolite complex. Solid symbols are projections on to the horizontal plane and open symbols are projections on to the vertical plane. Demagnetising temperatures are given in degrees centigrade or peak alternating fields in milli Teslas (mT).

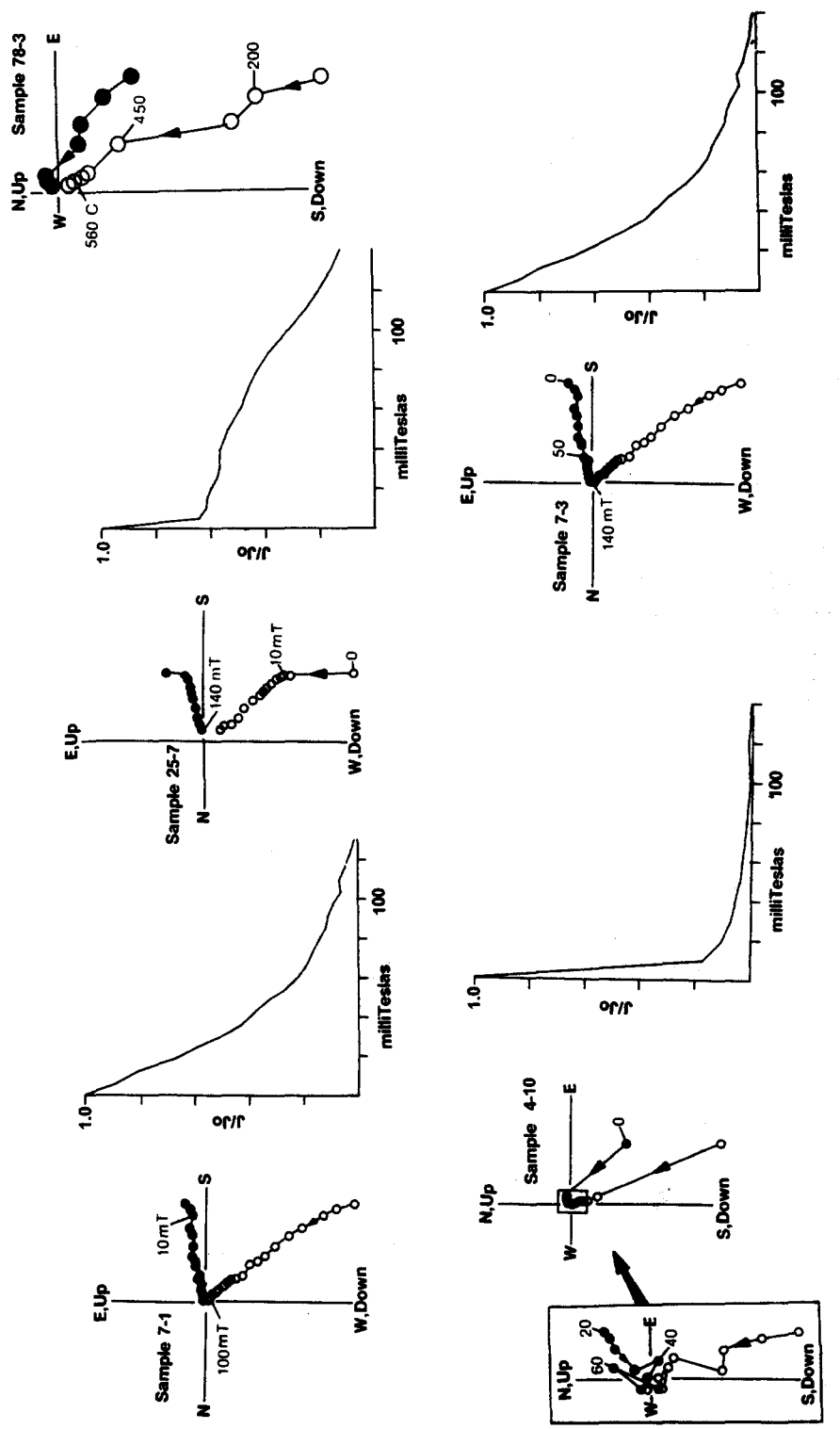


Fig. 7. Orthogonal vector projections and demagnetisation curves for typical components resolved by thermal cleaning belonging to the 'B' (SE+) group. Symbols are as for Fig. 6.

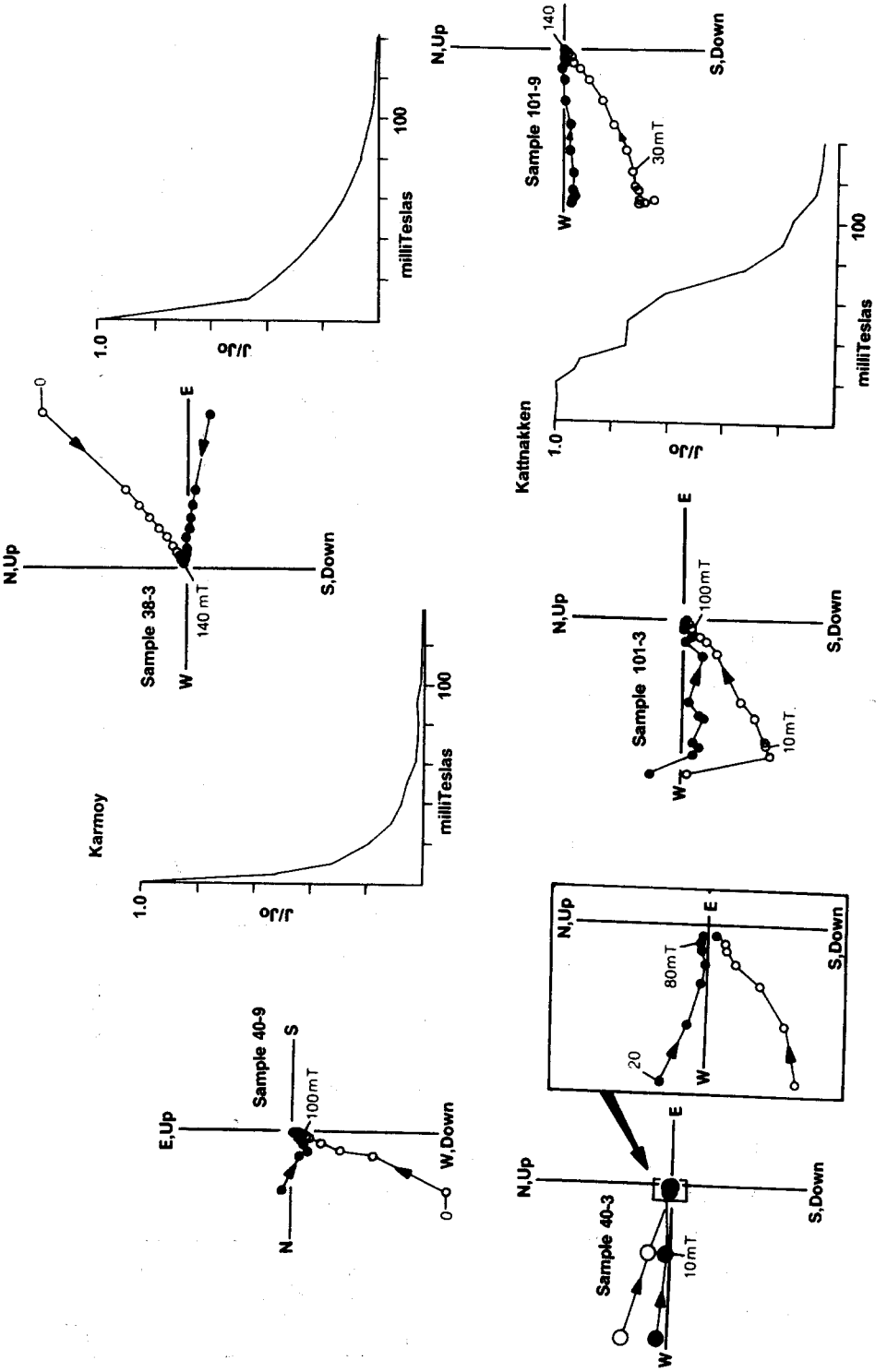


Fig. 8. Orthogonal vector projections and demagnetisation curves for typical components assigned to the 'C' (NNE+) and 'D' (W+) groups. Symbols are as for Fig. 6.

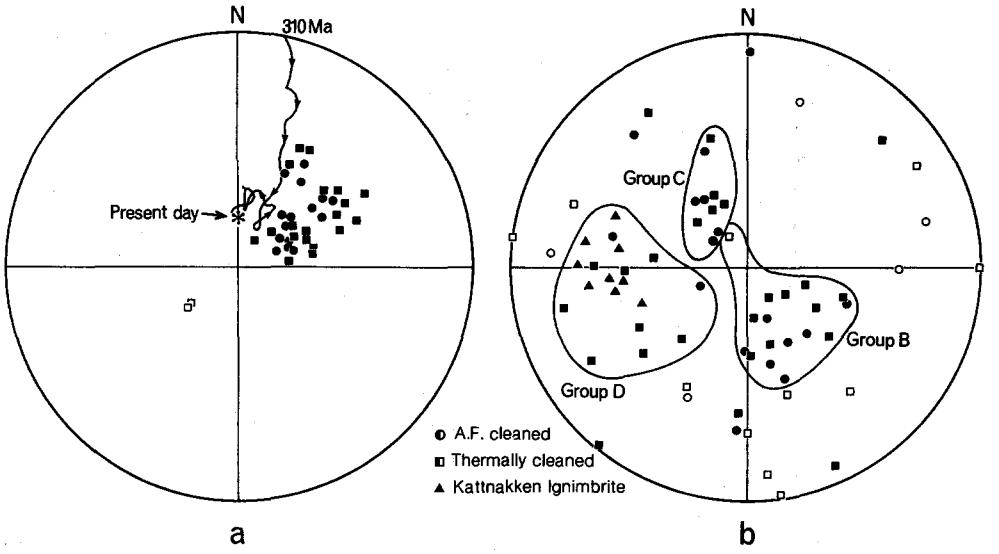


Fig. 9. Components of magnetisation identified by thermal and alternating field demagnetisation of samples from the Karmøy ophiolite complex and Kattnakken. (a) Components linked to the predominant NE+/SW- group identified by this study; the solid line is the predicted path of the palaeofield vector in this area since mid-Carboniferous times terminating in the present field (star); it is derived from the mean apparent polar wander path from the European plate (Piper, 1987). (b) All other components recognised by this study with groups of field directions (B–D) discussed in the text; three directions belonging to the 'C' group have been reversed for clarity.

This component yields an *in situ* group mean of $D = 270°$, $I = 38°$ with a palaeopole (D) at 286°E, 18°N (dp/dm = 4.6°/7.8° (Table 1)).

4. Interpretation of the palaeomagnetic results

A first step in interpretation of the components of magnetisation identified in these ophiolites must compare the remanence directions with the path of the palaeofield since the time of emplacement. There are still many uncertainties about the nature of APW during the interval spanned by the Caledonian orogeny (e.g., Briden et al. 1984; Piper, 1987), but the path for Eurasia is moderately well defined from mid-Carboniferous times onwards; the path of the palaeofield vector in southwest Norway computed from the mean APW path for Eurasia is shown in Fig. 9(a). None of the groups of directions from the ophiolites straddle this younger path in a way consistent with a simple post-mid-Carboniferous overprint origin: if they are post-Caledonian overprints they must have experienced deformation since magnetisation. The predominant A (NE+) component approaches closest to the palaeofield path in mid-Mesozoic times (~ 180 Ma), although all components plot to the east of the mean path.

There are few post-Caledonian igneous and metamorphic rocks in the Scandinavian Caledonides with which to assess the extent of younger overprinting in these facies. Palaeomagnetic results from younger dyke suites suggest that variable degrees of Mesozoic overprinting have occurred locally in Southern Norway. The Arendal dyke swarm in southeast Norway (Halvorsen, 1972) possesses shallow reversed magnetisations linked to the Permo-Carboniferous superchron ($D/I = 202°/26°$ and $214°/37°$); although a secondary component linked to Mesozoic events in the North Sea is present at some sites, it is of reversed polarity and removable by cleaning. To the northeast of Karmøy, the islands of Bømlo and Stord and regions to the northeast are cut by a number of widely spaced N–S-trending dolerite dykes. K–Ar data on these dykes are dispersed and they are variably considered to be between Carboniferous and Jurassic in age. Their predomi-

nant component of magnetisation (D/I = 48°/49°, Løvlie, 1981) plots near one margin of the 'A' components from Karmøy; some samples have a second reversed component of probable Permian age with the dual-component nature of the remanence being recognisable by progressive cleaning. On the island of Sotra west of Bergen and 120 km north of Karmøy, Løvlie and Mitchell (1982) identify a Jurassic remanence (D/I = 20°/59°) in a dyke swarm yielding consistent Permian (~250 Ma) K–Ar ages; they postulate that these dykes were completely remagnetised during a Kimmerian phase of uplift associated with North Sea rifting and uplift of this margin of Fennoscandia. Such an interpretation could only be applicable to the Karmøy 'A' components if remagnetisation has been succeeded by seaward tilting of this coastal margin. Adjustment for tilt about an axis parallel to the present coast (N350°E) is able to shift the declination towards the observed palaeofield, but also steepens the inclination; a seaward tilt of 20° could modify a vector of $D/I = 20°/73°$ approximating to the field at 210–170 Ma to yield the observed 'A' remanence. A similar mechanism has been invoked by Beckmann (1982) to explain the divergence of a remanence observed in presumed Lower Tertiary dykes from East Greenland from the predicted field; in this instance the required tilt is landward and attributed to loading by the regional ice cap. Unfortunately there are no coast-parallel dykes in the vicinity of Karmøy to test the efficacy of this mechanism, but the magnitude of the tilt required and the apparent absence of such a tilt on Sotra raise doubts about the validity of this interpretation. At Sotra the overprinting is linked to low-temperature ferri-rutile granulation (Løvlie and Mitchell, 1982); it would be very curious if such an event had influenced only the small number of mostly ultrabasic magnetically-stable lithologies and avoided all other rock types, including those apparently containing homogeneous magnetites (section 3 and Figs. 3 and 4). Furthermore the Mesozoic overprinting can plausibly be linked to permeating effects along young fractures, but the distribution of stable sites on Karmøy shows no apparent relationship to younger fractures and shear zones. The other major difficulty encountered by invoking Mesozoic tilting is that the inferred Permian component in nearby dykes accords with the Eurasian path in the in situ configuration (Løvlie, 1981).

The other possibility compatible with younger remagnetisation is that short-term variations in the Mesozoic APW path are suppressed by the mean path in Fig. 9 (averaged at 10-Ma intervals over time windows of 40 Ma) and that short-term deviations could accommodate the northnorthwest and east positive in situ directions of Fig. 9. While such an explanation might accommodate the (A) northeast + group, it is highly unlikely that such a short-term deviation could accommodate all of the magnetisations recognised by this study and notably the 'B' group (Fig. 9(b)) plotting remote from the mean post-310 Ma palaeofield path. Elsewhere in the circum-North sea regions, overprints linked to Mesozoic tectonism have been recognised by several investigations, (Storetvedt et al., 1978; Turner et al. 1978; Storetvedt and Carmichael, 1979; Torsvik et al., 1988). The poles from these studies straddle the palaeofield path (Fig. 10) and are not systematically aberrant like the results of this study.

In the absence of positive evidence for a post-Caledonian overprinting, we need to consider the case for remanence acquisition during the Caledonian orogeny and/or subsequent regional uplift and cooling. Several points are relevant to this assessment. (1) virtually all primary igneous assemblages in the ophiolite complex have been converted into low grade assemblages prior to, or during, obduction (Sturt et al., 1980); this metamorphism and related serpentinisation provides an older (mid-Ordovician) age limit for the remanence. (2) the gabbro and incorporated ultramafic lenses are strongly deformed; attempts to correct for this small-scale deformation randomises the coherent 'A' and 'B' groups: the remanence therefore post-dates the major (Finnmarkian, mid-Ordovician) folding in this area. (3) The strong magnetic properties of the pods are linked to the incidence of serpentinisation which post-dates this folding and the tectonic fabric (Sturt et al., 1979); this also requires that the magnetisation is a post-tectonic phenomenon. (4) The ultramafic bodies and other sites yielding these stable remanences

are all close to the intrusive contact with the WKIC and sited well within the likely thermal aureole; this would almost certainly have led to complete remagnetisation in Upper Ordovician (Ashgillian) times. Consistent with this observation, of the few stable components resolved from the WKIC the majority accord with the 'A' and 'B' group. (5) The mean direction adjusted for the tilt of the ophiolite yields a pole plotting to the west of Siluro-Devonian poles from Britain, but is otherwise unlike any known Ordovician pole for Europe. Since petrologic arguments also imply a post-emplacement remanence, there is therefore no case for considering the tilt-adjusted directions.

The 'A' group of northeast + components is both more tightly grouped and better represented within the ophiolite complex than the 'B' group of southeast + components. A declination of 90°E effectively divides the two groups (Fig. 9), although it is likely that they record a continuum of magnetising events during, or following, serpentinisation. This accompanied a regional rotation of ~100° with movement through palaeolatitude of no more than a few degrees (Table 1).

Thus we favour the view that the major components recognised in the study date from the post-emplacement, but Caledonian, history of the complex. If this interpretation is correct, the maximum estimate for the age of the poles is the timing of emplacement of the WKIC or ~445 Ma. A late Ordovician (Ashgillian) age would accord with a history of magnetisation pre-dating deposition of the overlying mid-Ashgillian–early Silurian Skudeneset supracrustals; by this time the complex had cooled and been elevated to surface levels.

The mean remanence direction from the Kattnaken ignimbrite ($D/I = 270°/38°$) is moved to $D/I = 305°/40°$ by adjustment for the tilt of the nappe complex. The in situ direction, however, lies close to directions widely resolved from Upper Silurian and Lower Devonian rocks in Britain, and notably within the Scottish sector of the Caledonides (Piper, 1987). We therefore suspect that it is essentially a post-folding overprint dating from uplift and cooling of the orogen. In addition, a number of west + components from Karmøy, which are somewhat more dispersed, lie in the same quadrant. Inclusion of these latter components yields a slightly more southerly direction ($D/I = 261°/46°$, Table 1). Although these inclinations are comparable, the declinations lie some 30–40° to the west of typical Siluro-Devonian results from Scotland and may therefore record a component of tectonic rotation about a vertical axis during the later Scandian (late Silurian–early Devonian) tectonic pulse.

5. Global implications

The results of this study merit comparison with the palaeomagnetic study by Taylor (1988) of the mid-Ordovician ophiolite at Unst in Shetland on the strike continuation of the belt in Scotland. A predominant 'A' component linked to post-emplacement cooling and dated ~435 Ma by the Ar^{39}/Ar^{40} method is comparable to the northeast + remanence at Karmøy ($D/I = 80°/68°$); two subsidiary components (yielding poles at 323°E, 20°S (dp/dm = 12.4°/13.6°) and 351°E, 35°S (dp/dm = 6.5°/16.7°)) are interpreted in the context of later events, including Devonian burial. The Unst 'A' pole plots between poles from the Karmøy 'A' and 'B' magnetisations and the collective mean of the 'A' and 'B' groups at Karmøy ($D/I = 74°/66°$) is nearly identical to the mean direction of the 'A' component from the Unst ophiolite ($D/I = 78°/68°$), although precise correlation of their declinations may not be realistic if the southwest Norway results have experienced some measure of clockwise rotation during Silurian times, as noted above. Nevertheless the general agreement between the Unst and Karmøy results, and especially the correspondence of their inclinations, is noteworthy for two reasons: (i) both complexes are linked to post-emplacement cooling in Middle–Upper Ordovician times; and (ii) the Unst ophiolite is emplaced on to the northwest (Laurentian) foreland of the Caledonian orogen whereas the Karmøy ophiolite is emplaced on to the southeast (Fennoscandian) foreland of the orogen. If the predominant components of magnetisation in each ophiolite are indeed of post-emplacement and uplift-related origin, this agree-

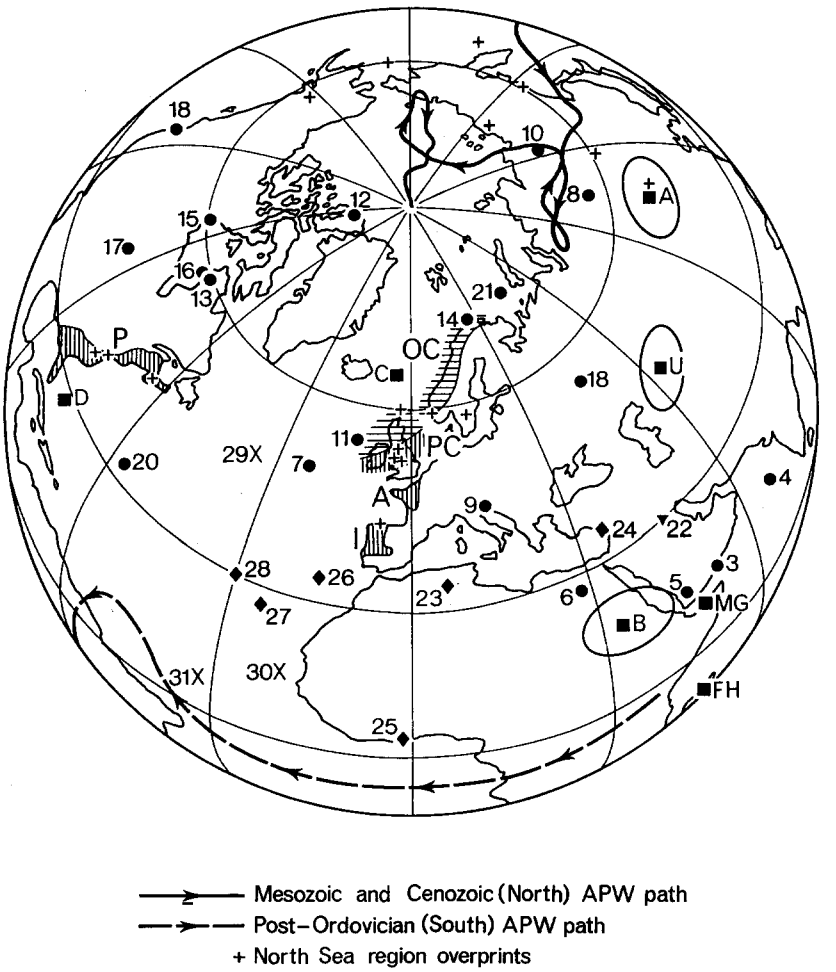


Fig. 10. Palaeomagnetic poles derived from rock units of known or probable mid-Upper Ordovician age in northwest Europe and the Piedmont of eastern North America. A, B, C and D are the four components identified from this study in southwest Norway, and U is the pole from the Unst ophiolite complex; these poles come from the orthotectonic Caledonides. Other poles are numbered as in Table II and come from the Baltic Platform (upright triangle), the paratectonic Caledonides (circles), Armorica (diamonds), Iberia (inverted triangles), corrected for post-Palaeozoic movements relative to Britain–Armorica, and the Piedmont of eastern North America (crosses, corrected for Phanerozoic opening of the North Atlantic). It is not yet clear how much of the dispersion between these data is to be attributed to apparent polar wander and how much is due to tectonic complexities, incomplete separation of magnetic components, and the fact that these terranes were not all welded fully together by these times (see text). The shaded APW path is the field of early Silurian to Lower Devonian palaeomagnetic poles from Britain (Piper, 1987). The large letters refer to: OC, orthotectonic Caledonides; PC, paratectonic Caledonides; A, Armorica; I, Iberia; and P, Piedmont. The post-Ordovician APW path is illustrated for comparison together with the poles of overprints from the North Sea region.

Notes to table 2:

[a] Compiled from Perroud and van der Voo (1985) and Piper (1987).

[b] Results 5–7 tilt corrected from original data from new geological evidence requiring that intrusions pre-date Ashgill folding.

[c] Probable Upper Ordovician age on petrologic grounds.

[d] a.f. Isolated component, thermal demagnetisation suggests that a shallow component is superimposed.

[e] Diagenetic remanence.

TABLE 2

Palaeomagnetic poles of possible upper Ordovician–early Silurian age [a]

Rock unit		Rock age	Pole coordinates		Palaeolatitudes	Reference
(1) *Orthotectonic Caledonides*						
Karmøy Ophiolite	A	Om–u, 445	106E	52N	38.6	This paper
	B		31E	17N	44.5	This paper
	C		244E	64N	44.5	This paper
Unst Ophiolite	A	435–425	64E	47N	64.2	Taylor (1988)
(2) *Paratectonic Caledonides*						
Snead Phacolith		Ou	53	12N	29.7	Piper (1978)
Corndon intrusion		Ou	69	15N	23.0	Piper (1978)
Shelve dolerites		Ou [b]	44E	12N	34.9	Piper (1978)
Moel-y-Golfa Andesite		Ou [b]	27E	28N	56.9	Piper and Stearn (1975)
Breidden Hill dolerites		Ou [b]	157E	50S	78.1	Piper and Stearn (1975)
Threlkeld–St. Johns Micro granodiorite		Ou, 438	102E	64N	42.1	Piper (1979)
Stockdale Rhyolite		Ou, 430	16E	45N	73.5	JDAP (unpublished)
WNW dykes, Cumbria		Ou	124E	68N	38.6	Piper et al. (1978)
E–W Kersantite dykes		Ou [c]	348E	54N	84.1	Piper et al. (1978)
Penmaenmawr granodiorite		Ou	104E	81S	55.5	Thomas and Briden (1976)
Penmaenbach rhyolite		Ou	112E	59S	55.5	Thomas and Briden (1976)
Foel Fras granodiorite		Ou	209E	71S	67.0	Thomas and Briden (1976)
Garnfor granodiorite		Ou	84E	60S	43.2	Thomas and Briden (1976)
Carreg-y-Llan granodiorite		Ou	111E	59S	55.5	Thomas and Briden (1976)
Bodeilias granodiorite		Ou	87E	45S	36.0	Thomas and Briden (1976)
Myndd Nefyn granodiorite		Ou	55E	44S	20.0	Thomas and Briden (1976)
Gurn Ddu granodiorite		Ou	233E	55S	58.5	Thomas and Briden (1976)
Yr Eifl microgranite		Ou	122E	32S	47.0	Thomas and Briden (1976)
Cader Idris basalts		Ou	229E	73S	58.5	Thomas and Briden (1976)
(3) *Baltic Platform*						
Orthoceras Limestones [d]		0	46E	30N	54.0	Claesson (1978)
(4) *Iberian–Armorican Massif*						
Thouars Massif		Ou, 444	5E	34N	76.4	Perroud and van der Voo (1985)
Erquy Spilites		Ou	34E	35N	60.1	Duff (1979)
Crozon dolerites		Ou	358E	3N	45.6	Perroud et al. (1983)
Moulin de Chateaupanne Formation		01 [e]	343E	34N	72.2	Perroud et al. (1986)
Bucaco red beds		01	335E	25N	69.6	Perroud and Bonhommet (1981)
Cabo de Penas volcanics		01	330E	30N	66.5	Perroud and Bonhommet (1981)
(5) *Appalachian sector*						
Delaware metamorphic rocks		Ou, 440	288E	48N	81.2	Brown and van der Voo (1983)
Arden Pluton		Ou, 440	303E	16N	61.1	Rao and van der Voo (1983)
Dunn Pt, Nova Scotia		Ou	300E	10N	53.5	Johnson and van der Voo (1983)

ment would support the view that the sector of the Iapetus Ocean between the Laurentian and Fennoscandian plates had effectively closed by Upper Ordovician times (Roberts and Gale, 1978).

The wider significance of these collective results is more difficult to evaluate because they are far removed from poles of Silurian and younger age (Fig. 10) and, indeed, from results from somewhat older units yielding poles inferred to be representative of Lower and Middle Ordovician times (Briden et al., 1984; Trench et al., 1988). Over the past decade a number of studies have identified magnetisations with steep positive or negative inclinations in rocks of known, or possible, Upper Ordovician age. Perroud and van der Voo (1985) have discussed palaeomagnetic evidence from some terranes now incorporated in the Caledonian–Acadian orogen yielding steep inclinations and implying a situation in high latitudes during Upper Ordovician times. Table 2 is an expanded compilation of these data incorporating results from the orthotectonic Caledonides and the Baltic Platform. Unfortunately, most of these rock units are in situations that have not permitted the proper application of palaeomagnetic field tests, and magnetic remanences cannot therefore be confirmed to be of Upper Ordovician age. In addition the extensive evidence from the paratectonic Caledonides is based mostly on a.f. demagnetisation studies and it is likely that the composition of the remanences will prove to be more complex when thermal demagnetisation is undertaken; the Orthoceras Limestones from the Baltic Platform for example, showed a migration of uncertain significance away from the a.f. defined direction when thermal cleaning was employed (Claesson, 1978). Similar problems may be associated with the Bucaco and Cabo de Penas results (Perroud and Bonhommet, 1981) from the Iberian Peninsula. The age of the rock units is not in all cases well defined; this reservation applies to most, but not all, of the intrusive suites from the paratectonic Caledonides and to the Crozon result from Armorica (Perroud et al., 1983). Nevertheless the poles from these Lower Palaeozoic rocks do not, in general, correspond to younger pole positions, and their dispersion contrasts with the picture that has emerged from younger Palaeozoic (Devonian–Permian) data from Europe (Piper, 1987).

There are several possible causes for this. One obvious reason is tectonic; not only may some of these terranes still have been separated by small oceans in Upper Ordovician times but there is also the possibility of regional and local rotations of the rock units during Caledonian and Acadian tectonic episodes following suturing; the possible Scandian (Silurian) effects at Karmøy are an example. This may be the reason why the dispersion of data within the terranes (1) to (5) in Table 2 is greater than the dispersion between terranes. Local tectonic deformations may take a variety of forms, with the resultant effect of yielding a dispersion of declinations and a distribution of pole positions along a small circle about the sample locality (Macdonald, 1980). This explanation is not, however, an adequate one for the data listed in Table 1, firstly because the data also show an approximately equal dispersion in inclinations (and palaeolatitudes), and secondly because individual units such as Karmøy yield a distribution of directions (and small circle distribution of poles), which is not fully explicable in terms of local tectonic effects. Thus the dispersion of 'A' and 'B' directions in the Karmøy complex is not related to individual rotations of small fault blocks and appears to record regional rotation of this sector of the Caledonides during uplift-related cooling.

The observation that the rock units of known or probable Upper Ordovician age listed in Table 2 yield intermediate to high palaeolatitudes suggests that these several terranes were close to each other by these times and presumably close to the northern margin of Gondwanaland, which was then being glaciated (Beuf et al., 1971; Piper, 1987). In addition, the data require that rapid apparent polar movements took place in Upper Ordovician and Lower Silurian times to move these regions back into low to intermediate palaeolatitudes defined by Silurian and Devonian data (Scotese et al., 1979). Since the Scottish–Scandinavia–East Greenland sector of the Caledonian orogenic belt had essentially closed up by

these times (cf. the Karmøy and Unst data and geological evidence (Roberts and Gale, 1978)), it is probable that high palaeolatitude palaeomagnetic results of Upper Ordovician age will in due course be recognised within the Laurentian block.

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